ARTIFICIAL INTELLIGENCE

Assignemnt 1

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Q1 Consider a map given in image.

Implement Breadth First Search(BFS) and Depth First Search (DFS) algorithm to find all possible step cost between Syracuse to Chicago.

BFS:

```
from collections import deque

def bfs_all_paths_costs(graph, start, goal):
    queue = deque()
    queue.append((start, [start], 0))   # (node, path, cost)
    results = []
    price=0

    while queue:
        current, path, cost = queue.popleft()

        if current == goal:
            results.append((path, cost))
            # don't break — we want *all* paths
```

```
        # expand neighbors breadth-first
        for neighbor, step_cost in graph[current].items():

            if neighbor not in path:  # avoid cycles
                price=price+ step_cost
                queue.append(( neighbor, path + [neighbor],price ))

    return results


# Define the graph
graph = {
    "Chicago": {"Detroit": 283, "Cleveland": 345, "Indianapolis": 182},
    "Detroit": {"Chicago": 283, "Cleveland": 169, "Buffalo": 256},
    "Cleveland": {"Detroit": 169, "Pittsburgh": 134, "Columbus": 144, "Chicago": 345, "Buffalo":
189},
    "Columbus": {"Cleveland": 144, "Indianapolis": 176, "Pittsburgh": 185},
    "Indianapolis": {"Chicago": 182, "Columbus": 176},
    "Pittsburgh": {"Cleveland": 134, "Buffalo": 215, "Philadelphia": 305, "Columbus": 185,
"Baltimore": 247},
    "Buffalo": {"Detroit": 256, "Pittsburgh": 215, "Syracuse": 150, "Cleveland": 189},
    "Syracuse": {"Buffalo": 150, "New York": 254, "Boston": 312, "Philadelphia": 253},
    "New York": {"Syracuse": 254, "Philadelphia": 97, "Boston": 215, "Providence": 181},
    "Philadelphia": {"New York": 97, "Pittsburgh": 305, "Baltimore": 101, "Syracuse": 253},
    "Baltimore": {"Philadelphia": 101,"Pittsburgh":247},
    "Boston": {"Syracuse": 312, "Providence": 50, "New York": 215, "Portland": 107},
    "Providence": {"Boston": 50, "New York": 181},
    "Portland": {"Boston": 107}
}

bfs_results = bfs_all_paths_costs(graph, "Syracuse", "Chicago")
count=0
print("\nBFS Paths & Costs:")
for path, cost in bfs_results:
    print(path, "=>", cost)
    count=count+1
print(count)
```

OUTPUT:

```
PS D:\SEM4\AI\LAB\Assignment1> python -u "d:\SEM4\AI\LAB\Assignment1\tempCodeRunnerFile.py"
BFS Paths & Costs:
['Syracuse', 'Buffalo', 'Detroit', 'Chicago'] => 3280
['Syracuse', 'Buffalo', 'Cleveland', 'Chicago'] => 5112
['Syracuse', 'Buffalo', 'Detroit', 'Cleveland', 'Chicago'] => 8758
['Syracuse', 'Buffalo', 'Pittsburgh', 'Cleveland', 'Chicago'] => 9416
['Syracuse', 'Buffalo', 'Cleveland', 'Detroit', 'Chicago'] => 10318
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Chicago'] => 14384
['Syracuse', 'Buffalo', 'Pittsburgh', 'Cleveland', 'Detroit', 'Chicago'] => 17889
['Syracuse', 'Buffalo', 'Pittsburgh', 'Columbus', 'Cleveland', 'Chicago'] => 19440
['Syracuse', 'Buffalo', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 19622
['Syracuse', 'Buffalo', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 20558
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Chicago'] => 21944
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Detroit', 'Chicago'] => 25256
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Chicago'] => 26692
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Chicago'] => 27519
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Chicago'] => 28033
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 28404
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Chicago'] => 29062
['Syracuse', 'Buffalo', 'Detroit', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 32100
['Syracuse', 'Buffalo', 'Pittsburgh', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 33192
['Syracuse', 'Buffalo', 'Pittsburgh', 'Columbus', 'Cleveland', 'Detroit', 'Chicago'] => 33858
['Syracuse', 'Buffalo', 'Cleveland', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 36101
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Detroit', 'Chicago'] => 37248
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Chicago'] => 38684
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Chicago'] => 39511
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Chicago'] => 40025
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 40396
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Chicago'] => 41054
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Chicago'] => 43694
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 45546
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago'] => 46261
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Cleveland', 'Chicago'] => 47277
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Detroit', 'Chicago'] => 47560
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Detroit', 'Chicago'] => 48484
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Detroit', 'Chicago'] => 50372
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Detroit', 'Chicago'] => 51808
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Chicago'] => 52635
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Cleveland', 'Chicago'] => 53149
```

```
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 53520
['Syracuse', 'Buffalo', 'Detroit', 'Cleveland', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 55945
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 60299
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago'] => 61014
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Cleveland', 'Chicago'] => 62030
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Detroit', 'Chicago'] => 62313
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Detroit', 'Chicago'] => 63237
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Detroit', 'Chicago'] => 65125
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Detroit', 'Chicago'] => 66561
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Chicago'] => 67388
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Cleveland', 'Chicago'] => 67902
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 68273
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Chicago'] => 68931
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Detroit', 'Chicago'] => 70702
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Chicago'] => 72138
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Chicago'] => 72965
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Chicago'] => 73479
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 73850
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Chicago'] => 74508
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 77145
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago'] => 78042
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 79189
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago'] => 79904
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Detroit', 'Cleveland', 'Chicago'] => 80920
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Detroit', 'Chicago'] => 81203
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Cleveland', 'Detroit', 'Chicago'] => 82127
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 905
81
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago'] => 91478
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 9
2625
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago'] => 93340
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Detroit', 'Cleveland', 'Chicago'] => 94356
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Detroit', 'Chicago'] => 94639
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Cleveland', 'Detroit', 'Chicago'] => 95563
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Detroit', 'Chicago'] => 97451
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Chicago'] => 98887
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Chicago'] => 99714
```

```
PS D:\SEM4\AI\LAB\Assignment1> python -u "d:\SEM4\AI\LAB\Assignment1\tempCodeRunnerFile.py"
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Chicago'] => 100228
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 100
599
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Chicago'] => 10125
7
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 1025
75
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago'] => 103290
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Cleveland', 'Chicago'] => 104306
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Detroit', 'Chicago'] => 104589
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Detroit', 'Chicago'] => 105513
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Cleveland', 'Detroit', 'Chicago'] => 107401
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Detroit', 'Chicago'] => 108837
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Chicago'] => 109664
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Cleveland', 'Chicago'] => 110178
['Syracuse', 'Boston', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Indianapolis', 'Chicago'] => 1105
49
['Syracuse', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 1114
83
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Columbus', 'Indianapolis', 'Chicago'] => 11
4873
['Syracuse', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago'] => 115770
['Syracuse', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Cleveland', 'Columbus', 'Indianapolis', 'Chica
go'] => 121804
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Columbus', 'Indianapolis', 'Chi
cago'] => 125194
['Syracuse', 'New York', 'Philadelphia', 'Baltimore', 'Pittsburgh', 'Columbus', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago'
] => 126091
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Columbus', 'Indianapolis', 'Chi
cago'] => 127238
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Cleveland', 'Buffalo', 'Detroit', 'Chicago']
 => 127953
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Detroit', 'Cleveland', 'Chicago']
 => 128969
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Buffalo', 'Cleveland', 'Detroit', 'Chicago']
 => 129252
['Syracuse', 'Boston', 'Providence', 'New York', 'Philadelphia', 'Pittsburgh', 'Columbus', 'Cleveland', 'Detroit', 'Chicago'
] => 130176
```

DFS:

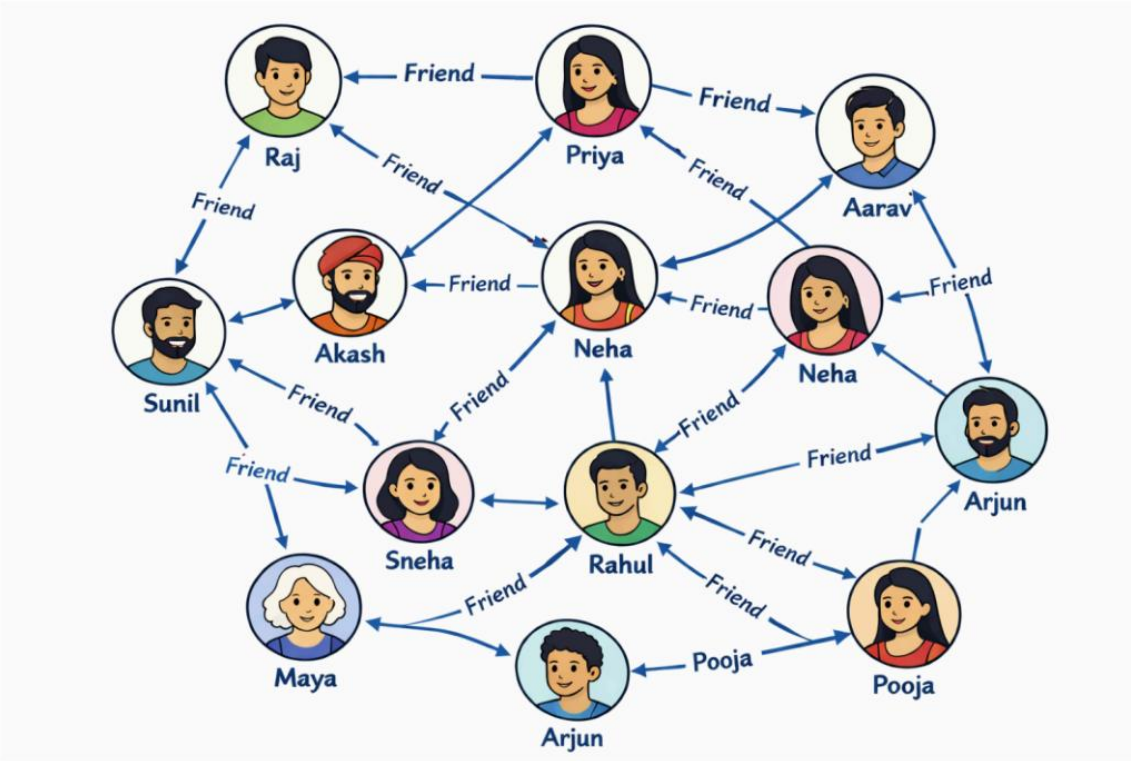
```
graph = {
    "Chicago": {"Detroit": 283, "Cleveland": 345, "Indianapolis": 182},
    "Detroit": {"Chicago": 283, "Cleveland": 169, "Buffalo": 256},
    "Cleveland": {"Detroit": 169, "Pittsburgh": 134, "Columbus": 144, "Chicago":
345},
    "Columbus": {"Cleveland": 144, "Indianapolis": 176, "Pittsburgh": 185},
    "Indianapolis": {"Chicago": 182, "Columbus": 176},
    "Pittsburgh": {"Cleveland": 134, "Buffalo": 215, "Philadelphia": 305, "Columbus":
185},
    "Buffalo": {"Detroit": 256, "Pittsburgh": 215, "Syracuse": 150},
    "Syracuse": {"Buffalo": 150, "New York": 254, "Boston": 312},
    "New York": {"Syracuse": 254, "Philadelphia": 97, "Boston": 215},
    "Philadelphia": {"New York": 97, "Pittsburgh": 305, "Baltimore": 101},
    "Baltimore": {"Philadelphia": 101},
    "Boston": {"Syracuse": 312, "Providence": 50, "New York": 215, "Portland": 107},
    "Providence": {"Boston": 50},
    "Portland": {"Boston": 107}
}
```

```
start = "Syracuse"
target = "Chicago"

#DFS
def stuck(x, seen): #Check if from city x, there is still a path to the target
    if x == target:
        return False
    for y in graph[x]:
        if y not in seen:
            seen.add(y)
            if not stuck(y, seen):
                return False
    return True

def search(path): #search for the path
    x = path[-1]
    if x == target:
        dis = 0
        for i in range(len(path) - 1):
            dis += graph[path[i]][path[i+1]]
        print(" -> ".join(path), f" Total Distance: {dis} miles")
        return

    seen = set(path)
    if stuck(x, set(seen)):
        return

    for y in graph[x]:
        if y not in path:
            path.append(y)
            search(path)
            path.pop()

print(f"All possible paths from {start} to {target}:\n")
search([start])
```

Output:

```
PS D:\SEM4\AI\LAB\Assignment1> python -u "d:\SEM4\AI\LAB\Assignment1\Q1.py"

Syracuse -> Buffalo -> Detroit -> Chicago  Total Distance: 689 miles
Syracuse -> Buffalo -> Detroit -> Cleveland -> Pittsburgh -> Columbus -> Indianapolis -> Chicago  Total Distance: 1252 miles
Syracuse -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago  Total Distance: 1077 miles
Syracuse -> Buffalo -> Detroit -> Cleveland -> Chicago  Total Distance: 920 miles
Syracuse -> Buffalo -> Pittsburgh -> Cleveland -> Detroit -> Chicago  Total Distance: 951 miles
Syracuse -> Buffalo -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago  Total Distance: 1001 miles
Syracuse -> Buffalo -> Pittsburgh -> Cleveland -> Chicago  Total Distance: 844 miles
Syracuse -> Buffalo -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago  Total Distance: 1146 miles
Syracuse -> Buffalo -> Pittsburgh -> Columbus -> Cleveland -> Chicago  Total Distance: 1039 miles
Syracuse -> Buffalo -> Pittsburgh -> Columbus -> Indianapolis -> Chicago  Total Distance: 908 miles
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Detroit -> Chicago  Total Distance: 1242 miles
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago  Total Distance: 1292
 miles
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago  Total Distance: 1135 miles
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago  Total Distance: 1410 miles
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis -> Chicago
  Total Distance: 1798 miles
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago  Total Distance: 1641 miles
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago  Total Distance: 1437 mile
s
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago  Total Distance: 1330 miles
Syracuse -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago  Total Distance: 1199 miles
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Detroit -> Chicago  Total Distance: 1515 miles
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Columbus -> Indianapolis -> Chicago  Total Dist
ance: 1565 miles
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Cleveland -> Chicago  Total Distance: 1408 miles
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Chicago  Total Distance: 1683 miles
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Columbus -> Indianapolis
-> Chicago  Total Distance: 2071 miles
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Buffalo -> Detroit -> Cleveland -> Chicago  Total Distance:
1914 miles
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Detroit -> Chicago  Total Distance:
 1710 miles
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Cleveland -> Chicago  Total Distance: 1603 miles
Syracuse -> Boston -> New York -> Philadelphia -> Pittsburgh -> Columbus -> Indianapolis -> Chicago  Total Distance: 1472 mi
les
```

Q2 Consider a social media network as given below. Find BFS and DFS tree.



Code:

BFS:

```
graph = {
    "Raj": ["Sunil", "Neha_1"],
    "Sunil": ["Raj", "Akash", "Sneha", "Maya"],
    "Akash": ["Sunil", "Priya"],
    "Priya": ["Raj", "Aarav", "Akash"],
    "Neha_1": ["Raj", "Akash", "Sneha", "Aarav"],
    "Sneha": ["Sunil", "Neha_1", "Rahul"],
    "Maya": ["Rahul", "Arjun_1", "Sunil"],
    "Aarav": ["Neha_1", "Arjun_2"],
    "Neha_2": ["Priya", "Neha_1", "Rahul", "Arjun_2"],
    "Rahul": ["Neha_1", "Neha_2", "Sneha", "Maya", "Pooja", "Arjun_2"],
    "Arjun_1": ["Maya", "Pooja"],
    "Arjun_2": ["Aarav", "Neha_2", "Rahul"],
    "Pooja": ["Arjun_1", "Arjun_2", "Rahul"]
}

from collections import deque
#BFS
def bfs_tree(graph, start):
    visited = set([start])
```

```
    parent = {start: None}
    queue = deque([start])

    while queue:
        node = queue.popleft()
        for neighbor in graph[node]:
            if neighbor not in visited:
                visited.add(neighbor)
                parent[neighbor] = node
                queue.append(neighbor)

    return parent

parents = bfs_tree(graph, "Raj")
print("BFS Tree:")

def print_tree(parent, node, level=0):
    print("  " * level + node)
    for child in parent:
        if parent[child] == node:
            print_tree(parent, child, level + 1)

print_tree(parents, "Raj")
```

OUTPUT:

```
PS D:\SEM4\AI\LAB\Assignment1> python -u "d:\SEM4\AI\LAB\Assignment1\Q2.py"
BFS Tree:
Raj
  Sunil
    Akash
      Priya
    Sneha
      Rahul
        Neha_2
        Pooja
    Maya
      Arjun_1
  Neha_1
    Aarav
      Arjun_2
```

DFS:

```
graph = {
    "Raj": ["Sunil", "Neha_1"],
    "Sunil": ["Raj", "Akash", "Sneha", "Maya"],
    "Akash": ["Sunil", "Priya"],
    "Priya": ["Raj", "Aarav", "Akash"],
```

```
    "Neha_1": ["Raj", "Akash", "Sneha", "Aarav"],
    "Sneha": ["Sunil", "Neha_1", "Rahul"],
    "Maya": ["Rahul", "Arjun_1", "Sunil"],
    "Aarav": ["Neha_1", "Arjun_2"],
    "Neha_2": ["Priya", "Neha_1", "Rahul", "Arjun_2"],
    "Rahul": ["Neha_1", "Neha_2", "Sneha", "Maya", "Pooja", "Arjun_2"],
    "Arjun_1": ["Maya", "Pooja"],
    "Arjun_2": ["Aarav", "Neha_2", "Rahul"],
    "Pooja": ["Arjun_1", "Arjun_2", "Rahul"]
}

def print_tree(parent, node, level=0):
    print("  " * level + node)
    for child in parent:
        if parent[child] == node:
            print_tree(parent, child, level + 1)

#DFS
def dfs_tree(graph, node, visited, parent):
    visited.add(node)
    for neighbor in graph[node]:
        if neighbor not in visited:
            parent[neighbor] = node
            dfs_tree(graph, neighbor, visited, parent)
    return parent

visited = set()
parent = {"Raj": None}

dfs_tree(graph, "Raj", visited, parent)

print("DFS Tree:")
print_tree(parent, "Raj")
```

```
PS D:\SEM4\AI\LAB\Assignment1> python -u "d:\SEM4\AI\LAB\Assignment1\Q2.py"
DFS Tree:
Raj
  Sunil
    Akash
      Priya
        Aarav
          Neha_1
            Sneha
              Rahul
                Neha_2
                  Arjun_2
                Maya
                  Arjun_1
                    Pooja
```